Response to Reviewers

Manuscript ID: 26-0767

"Simultaneous Contact Selection and Planning for Contact-Rich Manipulation with Cascaded Optimization"

We thank all three reviewers for their constructive comments and for highlighting the strengths of our work, including that SCSP addresses a real limitation of local contact-implicit model predictive control (CIMPC) methods (**R1**) and a real gap in existing CIMPC methods (**R2**), the proposed contact selection and ranking strategy is practical and effective (**R1**, **R3**), and that the experiments are thorough and extensive (**R3**) and above average for this literature (**R2**), with strong results across simulation and real-world hardware experiments without motion capture (**R1–R3**).

Reviewer 1

Q1: The CSO-as-MIQP claim is under-specified

Section III-D states that Eq. (32) is formulated as an MIQP, that the inner loop constitutes a QP, and that the overall CSO is an efficient MIQP. This is plausible only under additional assumptions that are not made explicit.

For a fixed sampled contact point, fixed contact geometry, fixed Jacobians, quadratic objective, polyhedral friction constraints, and an explicit active-set or binary-variable encoding of

$$\hat{\lambda}_{\text{env},i} = \max(A_i \hat{\lambda}_r + b_i, 0), \tag{1}$$

the problem can be written as a region-wise QP or as an MIQP. However, the paper does not provide the MIQP variables, big-M or indicator constraints, active-set constraints, or the precise quadratic form being solved. The implementation section also states that the CSO inner loop is solved using CasADi with SNOPT, which is an SQP/NLP solver rather than a QP or MIQP solver.

The paper should either present the exact QP/MIQP formulation and solver details, or weaken the claim to finite candidate search with local continuous force optimization under an SCM approximation.

Answer: Thank you for pointing out this inconsistency. We agree that our previous characterization of CSO as an MIQP, and of its inner problem as a QP, was imprecise.

In our actual implementation, the sampled contact location is not encoded as an integer decision variable, and we do not introduce binary variables, big-M constraints, indicator constraints, or use a mixed-integer solver. Instead, the discrete contact-location variable is handled explicitly through the finite candidate set $\mathcal{I}_{\text{avail}}$. For each candidate contact location, we solve a continuous optimization problem over the robot contact forces and then compare the resulting costs across candidates.

We need to correct an omission in our previous description. In addition to constructing a closed-form contact-dynamics approximation, SCM applies one global projection to the environment impulse. This projection introduces an additional coupled operation, so the resulting problem is even farther from a QP formulation. The previous implementation solved the continuous subproblem serially with SNOPT; the current implementation replaces this with GPU-parallel CEM. Overall, CSO is a lightweight optimization problem that permits efficient parallel solution, but it does not satisfy a QP formulation.

The revised formulation therefore does not introduce binary variables, big-M constraints, indicator constraints, or a mixed-integer solve. The timing table now labels this quantity "CEM cost time" because the implementation uses GPU-parallel CEM rather than a mixed-integer solver.

Q2: The SCM assumptions and error bound are too weak

The surrogate contact model relies on fixed object-environment contact sets and decoupled multi-contact dynamics. These are strong assumptions, especially during flipping, tipping, rolling, or simultaneous multi-contact interaction. Lemma 3 only gives a boundedness argument; it does not establish accuracy or useful dependence on contact geometry, active-set changes, force magnitude, or the off-diagonal blocks of the Delassus matrix.

The paper should state the regularity assumptions required for Lipschitz continuity of the LCP solution and provide empirical SCM error validation against a higher-fidelity contact solver.

Answer: Thank you for this important comment. We agree that SCM relies on strong assumptions and should include a more thorough analysis of modeling errors. Our motivation is that high-fidelity contact dynamics and complete contact detection are both nonsmooth and computationally expensive for optimization. Moreover, high-fidelity contact dynamics are highly geometry-dependent and involve nonsmooth contact transitions, which can introduce numerical instability during optimization. Therefore, the role of SCM is only to predict the correct object motion trend rather than provide high-fidelity contact simulation. In practical deployment, the influence of SCM approximation errors on SCSP is further alleviated by the high-frequency receding-horizon updates of the framework.

We need to correct our previous analysis. The error-bound lemma can become very loose when the Delassus matrix is rank deficient, because its smallest eigenvalue can be dominated by the artificial regularization term. In addition, a global Lipschitz-continuity statement for the LCP solution is difficult to justify when the contact set switches. We therefore removed the erroneous lemma and proof and replaced this theoretical claim with empirical contact-model evaluation.

We evaluate sliding, sticking, contact transition (including separation and new-contact establishment), and a reference-path flipping rollout in MuJoCo. The ground truth uses LCP contact dynamics, while contact points are obtained from MuJoCo collision checking. All models use $h = 0.002\,\text{s}$.

We compare MuJoCo's underlying QP model, ComFree, ContactSDF, and SCM, and report post-step linear- and angular-velocity errors as means, standard deviations, and P95 values in Table IV.

The results show that SCM is comparable to the other contact models overall: it has the lowest linear-velocity error in contact transition, remains close to the baselines in sliding and sticking, and is competitive on the reference-path rollout. We further generate ideal contact-wrench trajectories with each contact model and execute them in the manipulation evaluation. The ideal success rates are all high (0.91–0.99), showing that the generated wrench trajectories are generally feasible; higher-fidelity contact models do not automatically produce better deployed performance. In the direct-force evaluation reported in the manuscript's Table V, the three closed-form models perform strongly, with SCM achieving the best reported real success rate (0.88) and the best overall result. We therefore select SCM for CSO.

Q3: The ranking strategy appears internally inconsistent

The paper defines

$$\rho = \frac{\ell_{\text{near}} - \hat{\ell}_{\text{min}}}{\max(\hat{\ell}_{\text{max}} - \hat{\ell}_{\text{min}}, \epsilon)}. \tag{2}$$

With this definition, smaller ρ means that p_{near} is closer to the best candidate, and larger ρ means it is worse. However, Eq. (39) sets $\kappa_t = \mathbf{1}(\rho_t > \bar{\rho})$ and states that $\kappa_t = 1$ means p_{near} is sufficiently good. This reverses the intended meaning. Eq. (40) is also difficult to reconcile with Algorithm 2 and the described lifting/placing phases.

This section should be rewritten as a clear finite-state or hysteresis rule. As written, the trigger logic is not mathematically consistent.

Answer: Thank you for pointing out this inconsistency. We agree that the original definition of the ranking score was unclear and could lead to a reversed interpretation. In the revised manuscript, ρ_t is explicitly defined as the normalized ranking score of p_{near} based on its cost reduction $\Delta\ell_t^{\text{near}}$ and the range of cost reductions $[\Delta\ell_t^{\text{min}}, \Delta\ell_t^{\text{max}}]$ among all candidate contact locations:

$$\rho_t = \Pi_{[0,1]}\left(\frac{\Delta\ell_t^{\text{near}} - \Delta\ell_t^{\text{min}}}{\max(\Delta\ell_t^{\text{max}} - \Delta\ell_t^{\text{min}}, \epsilon)}\right) \tag{3}$$

It represents the relative ranking of the improvement achieved by p_{near} compared with other candidates. Therefore, a smaller ρ_t indicates that p_{near} provides less improvement and has a worse ranking, while a larger ρ_t indicates that p_{near} is more competitive.

We further clarify that the previous formulation using the absolute cost ℓ and the revised formulation using the cost reduction $\Delta\ell$ are equivalent under min-max normalization, except for the ordering convention. Specifically, since the cost reduction is defined as $\Delta\ell_t^i = \ell(\boldsymbol{x}_t) - \ell(\hat{\boldsymbol{x}}_t^+)$, we can obtain that

$$\rho_t = \Pi_{[0,1]}\left(\frac{\ell(\hat{\boldsymbol{x}}_t^{+,\text{min}}) - \ell(\hat{\boldsymbol{x}}_t^{+,\text{near}})}{\max(\ell(\hat{\boldsymbol{x}}_t^{+,\text{min}}) - \ell(\hat{\boldsymbol{x}}_t^{+,\text{max}}), \epsilon)}\right). \tag{4}$$

The above derivation shows that the previous formulation using the absolute cost ℓ was equivalent in principle but introduced ambiguity due to the reversed ordering convention. To avoid this confusion, we have rewritten the Ranking Strategy subsection in Section IV-B and reformulated ρ_t directly based on the cost reduction $\Delta\ell$. The revised manuscript also replaces the earlier counter-based switching rule with EMA smoothing of the binary trigger. To remove the previous smoothing strategy's dependence on contact estimation—in particular, the T_3 counter that counted consecutive contact steps—we use an EMA that is independent of contact-state estimates.

Q4: The CPO potential costs appear incorrectly specified

The lifting cost uses

$$\ell_{\text{obs}} = \log(\|p_{\text{ee}} - p_o\|^2 + \epsilon), \tag{5}$$

inside a distance threshold and minimizes it. This is not a repulsive potential unless the weight is negative, which is not stated. The placing cost says that ℓ_{con} attracts the end-effector to the reference contact point p_{ref}, but the formula uses p_o instead of p_{ref}:

$$\ell_{\text{con}} = \log(\|p_{\text{ee}} - p_o\|^2 + \epsilon). \tag{6}$$

This should be corrected or explained. A natural formulation would explicitly include a term such as $\|p_{\text{ee}} - p_{\text{ref}}\|^2$.

Answer: Thank you for identifying these inconsistencies. We have revised Section IV-C (highlighted in red). Specifically, the sign of the object repulsive potential ℓ_{obs} has been corrected to ensure a repulsive behavior, and the contact attraction term ℓ_{con} has been modified to attract the end-effector toward an inward offset from the reference contact point $\boldsymbol{p}_{\text{ref}}$ along its surface normal, with $\delta = 0.02\,\text{m}$. We also removed the previous attraction term toward the object geometric center $\boldsymbol{p}_o$. That term was intended to maintain sufficient contact under contact-estimation errors, but it could oppose the attraction toward $\boldsymbol{p}_{\text{ref}}$ and create a force balance that traps the optimizer in a local minimum. The revised placing cost is:

$$\ell_{\text{con}} = \|\boldsymbol{p}_{\text{ee}} - (\boldsymbol{p}_{\text{ref}} - \delta\boldsymbol{n}_{\text{ref}})\|^2. \tag{7}$$

This removes the conflicting center-attraction term and yielded good empirical performance in the revised experiments.

Q5: The result analysis does not isolate the proposed contributions

The experiments mainly show that the full SCSP system outperforms several baselines in aggregate success rate. They do not sufficiently explain which proposed mechanism causes which improvement. The causal link between the stated contributions and the results is therefore incomplete.

In particular, the paper should report evidence for each component:

- CSO: quality of selected contact priors, contact-location maps, correlation between CSO cost and execution success, and comparison against random, nearest, heuristic, and multi-start contact selection.

TABLE IV
POST-STEP VELOCITY ERRORS RELATIVE TO LCP (MEAN ± STD AND P95). LINEAR: m/s; ANGULAR: rad/s. BOLD INDICATES MINIMA.

Contact state	MuJoCo QP				ComFree				ContactSDF				**SCM**			
	Linear	Angular	P95 Linear	P95 Angular	Linear	Angular	P95 Linear	P95 Angular	Linear	Angular	P95 Linear	P95 Angular	Linear	Angular	P95 Linear	P95 Angular
Contact transition	0.73 ± 0.27	$\mathbf{4.41 \pm 4.72}$	1.19	16.33	0.77 ± 0.27	4.78 ± 4.44	1.19	15.44	0.36 ± 0.16	9.44 ± 5.04	**0.65**	18.27	$\mathbf{0.26 \pm 0.21}$	6.53 ± 4.41	0.78	15.84
Sliding	0.30 ± 0.17	5.23 ± 4.14	0.69	15.59	0.34 ± 0.17	5.80 ± 4.38	0.71	16.66	$\mathbf{0.12 \pm 0.10}$	$\mathbf{3.64 \pm 2.84}$	**0.33**	**9.16**	0.15 ± 0.16	4.27 ± 3.85	0.52	13.58
Sticking	0.16 ± 0.21	1.92 ± 2.72	0.63	7.91	0.17 ± 0.20	1.92 ± 2.71	0.64	7.92	$\mathbf{0.09 \pm 0.11}$	$\mathbf{1.28 \pm 1.75}$	**0.36**	**5.53**	0.16 ± 0.21	1.98 ± 2.70	0.64	7.96
Reference-path rollout	0.03 ± 0.10	0.47 ± 1.55	0.21	3.22	0.03 ± 0.11	0.47 ± 1.53	0.24	3.18	$\mathbf{0.02 \pm 0.05}$	0.34 ± 1.13	**0.09**	2.22	0.02 ± 0.08	$\mathbf{0.29 \pm 1.03}$	0.12	**1.70**

- SCM: prediction error against a higher-fidelity contact solver, separated by contact count, contact mode, object geometry, and flipping/rotation phase.
- Ranking strategy: switch frequency, time spent tracking $\hat{p}_r^*$ versus p_{near}, failure cases caused by switching, and sensitivity to $\bar{\rho}, T_1, T_2, T_3$.
- CPO objective: ablation of lift, place, object-center attraction, and p_{ref} tracking terms.
- Robustness: failure categories separating perception error, contact estimation error, slippage, control saturation, and poor prior selection.

Without these analyses, claims about robustness, diversity, and the specific benefit of each module remain qualitative.

Answer: For CSO, we directly apply the optimized contact wrench at the selected contact and roll out the manipulation. The direct-force results in Table V achieve nearly 100% ideal success for SCM, which supports the quality of the selected contact priors. Figure 9 visualizes the contact maps for the elephant and bunny flipping tasks, including the CSO-optimal contact p_r^*, the nearest contact p_{near}, and the physical contact observed during execution. In Section V-B, "Functional Analysis of SCSP Components," A1 follows the optimal CSO contact, A2 always uses the nearest contact, and A3 uses a coarse heuristic contact model; these results isolate optimal, nearest, and heuristic selection. We did not add separate random and multi-start baselines: CSO already evaluates the candidate set in parallel, which provides the multi-start computation pattern, while purely random contacts are too unreliable to yield a meaningful executable comparison.

For SCM, Table IV reports linear- and angular-velocity errors and P95 values for contact transition, sliding, sticking, and the reference-path flipping rollout. The benchmark uses multi-contact configurations, seven object meshes with varied irregular geometries used in the experiments, and an LCP reference with MuJoCo collision checking. The direct-force rollout results in Table V further test whether the predicted contact wrenches are executable; the high ideal success rates across models show that the trajectories are generally feasible, while the deployed results show that higher model fidelity alone does not guarantee better task performance.

For the ranking strategy, Table VI reports the CSO-optimal-point switching frequency and physical contact-switching frequency. Table II compares execution time for SCSP with A1 and A2, showing the efficiency difference between ranking-based execution, strict optimal-contact tracking, and nearest-contact execution. Table VI also reports the normalized advantage of an individual ranking decision, computed from a time-discounted success reward, thereby measuring its effect on execution efficiency and outcome. Table VIII reports failure categories and recovery rates for real hardware, including slippage and failures associated with contact switching. Finally, Fig. 10 evaluates sensitivity to the acceptance threshold $\bar{\rho}$ and EMA coefficient β. The earlier T_1, T_2, T_3 counter-based smoothing is replaced by contact-estimation-independent EMA smoothing.

For CPO, A6 removes the lift-and-place cost and directly solves an obstacle-avoidance trajectory-optimization problem, while A7 replaces the CPO objective with an MPPI-based CIMPC objective. In the current formulation, the contact target is an inward-offset (penetrating) point $\boldsymbol{p}_{\text{ref}} - \delta \boldsymbol{n}_{\text{ref}}$, rather than an attraction term toward the object center. In the real-world ablation A9, we replace this inward-offset target with the estimated surface point $\boldsymbol{p}_{\text{ref}}$; this surface-attraction design often prevents the robot from descending to establish contact. Table VIII reports the resulting contact-estimation-related errors together with the success and recovery outcomes. The same table counts perception error, contact-estimation error, slippage, control saturation, and poor prior selection, and reports the recovery rate after each event. An event does not necessarily imply final failure because SCSP replanning can sometimes correct it; the recovery rate therefore counts the trials that still succeed after the event. Together with the CSO, SCM, ranking, and hardware analyses above, these experiments quantify the proposed components within the tested tasks.

Q6: Baseline coverage and comparison protocols need strengthening

The Sampling comparison is not fully controlled because the paper replaces the original contact model with f_{cf}. This may create a new baseline rather than a direct implementation of the cited method. The DyWA comparison also needs careful framing: DyWA is described in Table I as lacking generalization, although its stated purpose is generalizable nonprehensile manipulation. The paper should define exactly what "generalization" means.

The baseline set is also incomplete relative to the scope of the claims. The following comparisons or discussions are particularly relevant:

- STOCS / STOCS-3D: this is the closest prior work in spirit, since it directly addresses simultaneous trajectory optimization and contact selection with high-fidelity geometry. Even if it is not real-time, it should be used as an offline planning-quality reference or excluded with a clear justification.

- ContactSDF: this should be discussed for the paper's multi-contact, differentiable contact-model, and dexterous-manipulation claims. If a direct comparison is not feasible, the paper should explain the domain difference and narrow the multi-contact claims.
- Generic sampling-based optimizers: the paper should compare against MPPI, CEM, CMA-ES, or random shooting under the same dynamics model, cost, action parameterization, horizon, and wall-clock budget. Otherwise it is unclear whether SCSP's improvement comes from its contact-selection structure or simply from spending computation on sampled contact/control candidates.

A useful protocol would include both contact-location sampling baselines, e.g., CEM/CMA-ES over p_r and λ_r, and control-sequence sampling baselines, e.g., MPPI/CMA-ES over end-effector waypoints or joint commands without CSO. These should be run with the same model and compute budget as SCSP.

Answer: We thank the reviewer for the valuable suggestions. We have revised the baseline implementation and clarified the comparison protocols in the revised manuscript.

We now fully reproduce the released Sampling implementation, including its C3 contact model, the MIQP solved with Gurobi, and its original contact-selection strategy. The updated comparison results are reported in Table II.

Our use of "generalization" refers to cross-task generalization, whereas DyWA's reported setting concerns cross-object generalization. We have revised the wording in Table I to make this distinction explicit.

We additionally include STOCS-3D and ContactSDF in the comparison. STOCS-3D is a CITO method that cannot be evaluated online in our setting and does not switch contact points during execution. Given the initial and target object poses, we compute one contact trajectory offline with STOCS-3D and execute it through OSC. ContactSDF provides a differentiable contact model but does not perform explicit contact selection. The resulting comparison is reported in Table II.

We also evaluate direct sampling-based alternatives. A5 uses CMA-ES to optimize the object-surface contact point directly, with the SDF distance-to-surface constraint $d_{\mathrm{SDF}} = 0$. A7 directly explores end-effector displacement sequences with MPPI under the same contact model f_{cf}, control bounds, and cost weights as CPO ($N = 10$, 128 samples, one iteration, noise $\sigma = 0.003\,\mathrm{m}$, temperature 0.005), without any CSO output; the attraction term targets the object center. The results are included in Table II. A5 remains below SCSP, especially on objects with many sharp edges such as the bunny, where direct surface optimization often selects difficult contacts. Although MPPI explores locally more effectively than gradient-based methods such as CF-MPC, it still has difficulty discovering large-range contact switching, so its success rate remains limited.

Q7: Multi-contact and generalization claims are overstated

The notation allows multiple robot contact points, and the SCM includes multiple environment contacts. However, the main nonprehensile manipulation experiments use point-contact end-effectors, and the SCM explicitly neglects coupling between environment contacts. The later soccer, grasping, and long-horizon experiments are useful case studies, but they do not establish a general simultaneous multi-contact planning capability.

The claim that SCSP can be easily extended to multi-contact tasks should be weakened. The paper should discuss scalability with respect to the number of robot contacts, since selecting n_r contacts from n_s candidates can scale combinatorially, e.g., $\binom{n_s}{n_r}$ in the unordered case.

Answer: Thank you for pointing out this important limitation. We agree that extending SCSP to general multi-contact tasks faces scalability challenges. In Section VI-A(3), we demonstrate that SCSP can autonomously select finger contacts and generate force-closure grasps without manually specifying contact locations or the number of fingers. By leveraging GWB-based methods to reduce the CSO search space, this result demonstrates the potential of SCSP for extending contact selection and planning to multi-contact manipulation.

However, we acknowledge that more complex multi-contact tasks involve not only combinatorial growth but also strong geometric coupling among contact locations, which remain challenging for the current SCSP framework. The demonstrated precision grasping case involves a limited number of contacts with relatively weak coupling among fingers, while more complex tasks such as power grasping remain challenging for autonomous planning. We have therefore weakened our claims in the revised manuscript, stating that SCSP provides a potential framework for multi-contact manipulation rather than a general solution. This limitation and the scalability of multi-contact planning are also discussed as future work in Section VII.

Q8: Hardware evidence is encouraging but statistically limited

The real-world experiments use only five trials per object/surface condition. This is too small to support strong robustness claims. The paper should either increase the number of trials or report confidence intervals and failure categories. It should also report how often failures come from perception, contact estimation, slippage, control saturation, or poor contact prior selection.

Answer: Thank you for pointing out this limitation. Across the smooth and rough surface conditions, we test five task types—rotation of the bunny, rotation of the duck, flipping of the bunny, flipping of the hand, and flipping of the Xbox controller—with 30 trials for each task/surface condition, for 300 real-world trials in total. Table VII reports the success rates and two-sided Wilson 95% confidence intervals.

We also classify unsuccessful trials by their terminal outcomes, including timeout and falling off the designated workspace/table. For perception error, contact-estimation error, slippage, control saturation, and poor contact-prior selection, an individual event does not necessarily cause final failure because SCSP replanning can sometimes recover. Therefore, Table VIII reports both the number of occurrences and the

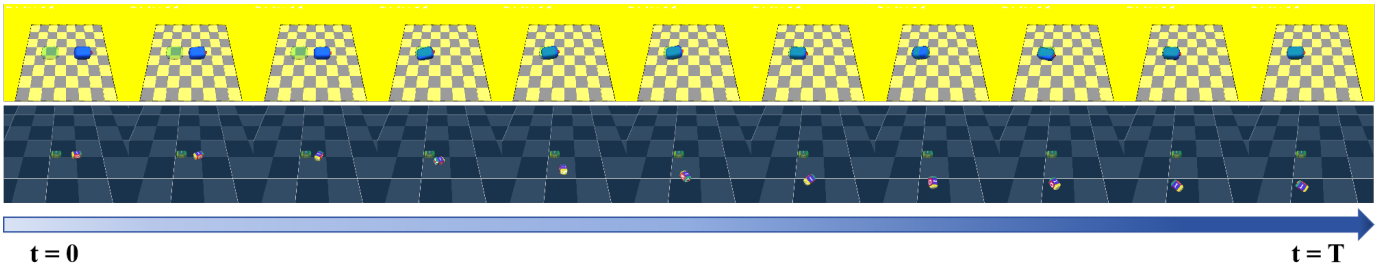


Fig. 1. Comparison between the trajectory planned by STOCS-3D (top) and its execution through contact-force tracking in MuJoCo (bottom). Although STOCS-3D produces a reasonable planned trajectory, the objec gradually deviates from the predicted trajectory during execution due to the mismatch between the quasi-dynamic planning model and the full contact dynamics in MuJoCo.

recovery rate, namely the fraction of trials that still succeed after the corresponding event.

Q9: Technical Corrections

- The friction constraints in Eq. (32) should include both signs of each tangential component, e.g.,

$$|\hat{\lambda}^{t1}_{r,i}| \le \mu_i \hat{\lambda}^{n}_{r,i}, \qquad |\hat{\lambda}^{t2}_{r,i}| \le \mu_i \hat{\lambda}^{n}_{r,i}. \tag{8}$$

- SCM is called "smooth," but the max operator is piecewise linear and nonsmooth at the switching surface.
- Eq. (35) should be rechecked for units and consistency between force and impulse notation.
- The text refers to the optimal contact location as $\hat{\lambda}^*_r$ in one place; this should be $\hat{p}^*_r$.
- The quaternion error in Table II is labeled as radians, but the reported metric appears to be $1 - (\psi^T \psi_g)^2$, not an angular error in radians.
- Lemma 4 is not a proof of non-conflicting objectives. It should be removed, weakened, or replaced with a precise first-order condition.

Answer: We thank the reviewer for the careful checking of the technical details. We have revised Eq. (32) and Eq. (35) to ensure the correctness of the friction constraints and the consistency of force/impulse notation. We have also corrected the notation error after Eq.(32) by replacing $\hat{\boldsymbol{\lambda}}^*_r$ with $\hat{\boldsymbol{p}}^*_r$, removed the incorrect radian unit annotation in Table II, and clarified that the quaternion error metric is a dimensionless quantity.

~~The current manuscript instead implements softplus$_T$ and describes the fixed-contact map as piecewise smooth and closed-form, with smoothness qualified by clipping and zero-impulse boundaries.~~ We agree that the max operator is nonsmooth at the switching surface. The current manuscript replaces it with the componentwise smooth approximation $\mathrm{softplus}_T(x) = T\log(1 + e^{x/T})$ and applies a single smooth residual correction to the diagonal impulse estimate instead of hard clipping. Lemma 2 is restated accordingly: for a fixed contact set, the SCM map is smooth and closed-form in $\boldsymbol{\lambda}_r$, and nonsmoothness arises only when the contact set changes.

We agree that the previous argument was not rigorous enough to support the claimed result. Lemma 4 and its proof have therefore been deleted from the current manuscript.

Reviewer 3

Q1: Contradictory ranking-strategy definitions

The ranking strategy contains contradictory definitions as printed. In (38), ρ is normalized so $\rho \sim 0$ means p_{near} is close to the best candidate, and (39) sets $\kappa = 1$ when $\rho > 0.75$, i.e., $\kappa = 1$ marks p_{near} as a poor candidate. Under this reading, (40), Fig. 6, and Algorithm 2 are mutually consistent (infeasible $\to$ $\gamma = 0$ $\to$ lift toward the CSO optimum). But the sentence directly below (39) states the opposite, that $\kappa = 1$ indicates p_{near} is "sufficiently good," and the sentence introducing the trigger defines "a trigger variable γ" and then writes the formula for κ. In addition, both branches of (40) gate on the same counter $t_{(\kappa=1)}$, defined after Algorithm 2 as the total timesteps satisfying $\kappa = 1$; I suspect the first branch should use $t_{(\kappa=0)}$, and it is not stated whether the counters are cumulative or consecutive, or when they reset. As written, the placing branch triggers only after T_1 accumulated infeasible steps, which does not match the described behavior. Please correct the prose and (40) from the implementation and define the counter semantics; at present the switching logic cannot be implemented unambiguously.

Answer: Thank you for the careful observation. We agree that the original formulation contained inconsistencies between the definition of ρ, the interpretation of κ, and the switching logic. The ambiguity mainly originated from a derivation error when simplifying the original formulation.

In the revised manuscript, we have rewritten Section IV-B and reformulated ρ_t with a clearer definition. Specifically, for each candidate contact location $p_i \in \mathcal{I}_{\text{avail}}$, the optimal contact force $\boldsymbol{\lambda}^*_i$ is solved to obtain the corresponding optimal cost reduction $\Delta\ell^i_t = \ell(\boldsymbol{x}_t) - \ell(\hat{\boldsymbol{x}}^+_t)$. The ranking score ρ_t then measures the relative ranking of the cost reduction obtained by p_{near} among all candidate contact locations. Therefore, ρ_t is reformulated as follows:

$$\rho_t = \Pi_{[0,1]}\left(\frac{\Delta\ell^{\text{near}}_t - \Delta\ell^{\min}_t}{\max(\Delta\ell^{\max}_t - \Delta\ell^{\min}_t, \epsilon)}\right). \tag{9}$$

With this definition, a larger ρ_t indicates that p_{near} provides greater improvement and has a better ranking. Accordingly, $\kappa_t = \mathbf{1}(\rho_t \ge \bar{\rho})$ consistently indicates whether p_{near} is sufficiently good. The previous derivation error was introduced during a further simplification of the formulation, as shown in (4).

The current manuscript replaces the counter-based rule with exponential moving-average smoothing, $\gamma_{t+1} = (1-\beta)\gamma_t + \beta\kappa_t$, with $\gamma_0 = 0$. The earlier smoothing still depended on contact information, in particular the T_3 counter of consecutive contact steps, which is not robust to contact-estimation errors in the real-world experiments. The EMA does not require any contact-related input and reduces the tuning burden by replacing the multiple counter thresholds with the single smoothing coefficient β. Its feasibility is evaluated in the real-world experiments in Section V-C and in the sensitivity study in Fig. 10.

Q2: Optimization claims

CSO performs a global search over the sampled candidate set, not over the continuous surface, so "globally optimal" (Fig. 2 caption) and especially "asymptotically optimal" (after (32), where the symbol should also be $\hat{p}_r$, not $\hat{\lambda}_r$) are overclaims; please reword or prove. The paper calls the problem an MIQP with sub-millisecond QP subproblems, but the implementation uses SNOPT, an SQP solver for NLPs, and with the quaternion cost the subproblem is not a QP. Algorithm 1 is titled "Sampling and Exchange Method" and Table V lists an "Exchange cost time," but no exchange step is defined anywhere. Since 70×0.23 ms is about the reported 16 ms CSO time, I assume the candidates are simply enumerated; if so, just say so.

Answer: Thank you for pointing out these inconsistencies.

We agree that "asymptotically optimal" was incorrect. CSO searches only the finite sampled candidate set, so the selected contact is optimal among the sampled candidates rather than asymptotically optimal on the continuous object surface. We have changed the wording accordingly and corrected the variable error in Eq. (32) from $\hat{\boldsymbol{\lambda}}_r^*$ to $\hat{\boldsymbol{p}}_r^*$.

The MIQP/QP characterization was also inaccurate. ~~The SCM relation contains a componentwise max, so the underlying force map is piecewise smooth and nonsmooth at activation boundaries rather than a QP.~~ The previous SCM relation contained a componentwise $\max$; the current manuscript replaces it with $\mathrm{softplus}_T$, so the force map is smooth for a fixed contact set, but the nonlinear softplus and the quaternion cost still prevent a QP formulation. In addition, the previous manuscript omitted the one-step global projection of the environment impulse; this extra coupled operation makes the problem even less compatible with a QP formulation. Nevertheless, the resulting optimization remains lightweight because the model is ~~piecewise smooth~~smooth and closed-form for a fixed contact set and the candidate set is finite.

The previous implementation solved the continuous subproblem serially with SNOPT, which was already sufficiently fast for the tested setting. The current implementation replaces this serial solve with GPU-parallel CEM and obtains higher solving efficiency. No binary variables, big-M constraints, or mixed-integer solver are used.

~~Algorithm 1 currently describes offline FPS followed by online valid-region selection; the table still reports an "Exchange cost time" entry, but no separate exchange optimization step is defined.~~ Algorithm 1 describes offline FPS followed by online valid-region selection. We have defined the Exchange step explicitly in Section III-B: it applies the offline-specified rule-based and manual masks online and exchanges the convex approximation $\mathcal{F}$ for the retained valid surface $\mathcal{F}_{\mathrm{v}}$, from which $\mathcal{I}_{\mathrm{avail}}$ is selected. The "Exchange cost time" entry in the computation-time table reports the cost of this masking step; it is not a separate optimization. The candidates are then simply enumerated, as the reviewer surmised.

Q3: Text vs. tables

Sec. V-C states SCSP achieved a 100% success rate in on-ground rotation, but Table IV shows 19/20 (Duck, rough surface: 4/5). The open-loop numbers for the long-horizon tasks are missing from Table VI although the text calls the rates "comparable." Tables IV and VI say "(%)" but list raw counts. "Significantly outperforms" is not accurate for multi-finger grasping (16/20 vs 15/20). Please reconcile all of these. I would also suggest confidence intervals: with n = 200 they are trivial to add, and with n = 5 per hardware cell the current wording is stronger than the data supports.

Answer: Thank you for the suggestions.

We have corrected all of these inconsistencies. The real-world evaluation now uses 30 trials for each object/surface condition, and the revised success rates and two-sided Wilson 95% confidence intervals are reported in Table VII. The open-loop long-horizon results are included in the success-trial table, and the wording for multi-finger grasping has been corrected to avoid overstating the comparison.

Q4: Reproducibility

The friction-cone constraints in (32) have no lower bounds on the tangential components as printed (and n_t vs n_d is inconsistent with (9)); please print the exact constraints used. Values for w_{att}, w_{obs}, w_{con}, w_o (is this the same as w_{obj}?), σ, the lift offset d, and solver settings are missing. Please also mark the 4.82 s sampling in Table V as offline so the real-time claim is unambiguous.

Answer: Thank you for pointing out these issues. ~~We have added the concrete parameter settings to the experimental-setting description, including the exact friction constraints, w_{att}, w_{obs}, w_{con}, w_{obj}, σ, the lift offset d, and the solver settings.~~ We have added the concrete settings to the experimental-setting description in Section V. The exact friction constraints are now printed in the CSO formulation, $|\hat{\lambda}^{t1}_{r,i}| \le \mu_i \hat{\lambda}^n_{r,i}$ and $|\hat{\lambda}^{t2}_{r,i}| \le \mu_i \hat{\lambda}^n_{r,i}$ with $0 \le \hat{\lambda}^n_{r,i} \le \hat{\lambda}^n_{\max}$. The lifting and reference-contact attraction weights are $w_{\mathrm{att}} = w_{\mathrm{con}} = 20$. The repulsive-field term is disabled in the reported experiments, so w_{obs} and σ are not used; the previous object-center attraction term and its weight w_{obj} have been removed from the CPO cost (see our reply to Reviewer 1, Q4). The single lift offset d is replaced by the lift–cross–drop path with clearance margins $d_{\mathrm{top}} = 0.025$ m and $d_{\mathrm{ref}} = 0.015$ m. The solver settings are also stated: CSO uses GPU-parallel CEM with 20 iterations over $n_s = 70$ candidates, and CPO uses `acados` SQP-RTI with HPIPM and a tolerance of 10^{-4}, with $N = 5$, $|u_j| \le 0.005$ m, $w_{\mathrm{lp}} = 1$, $w_u = 50$, and $w_{\mathrm{cso}} = 10$. We also mark the 4.82 s sampling cost as offline in the computation-time table.

Q5: Minor comments

In (33), α is introduced as a weighting coefficient but never used. $\tilde{Q}$ appears in (8) and (11) but only Q is defined in (4). In (47), the terminal term is $w_{\text{cso}}\ell_{\text{cso}}$ while the text says V_f as in (7); the same equation mixes t and k as time indices. In (13) and (32) the objective carries a free index k; presumably x_T is meant. (18) is undefined when n_i is parallel to the reference axis a; please state the fallback. Please state the orientation tolerance in (51) in degrees (about 26 deg) and the distribution behind "mass randomly scaled within a factor of 10."

Answer: Thank you for pointing out these issues. The $\tilde{\boldsymbol{Q}}$ in (8) and (11) is a notation error and should be $\boldsymbol{Q}$ as defined in (4). We have corrected this inconsistency in the revised manuscript.

The current manuscript retains the piecewise ℓ_{cso} definition and uses $\boldsymbol{x}_T$ in the CSO objective; it does not claim the separate unified-objective rewrite described in the previous response. ~~The coefficient α is still mentioned in the stable-pose prose but is not used in the displayed formula.~~ The unused coefficient α has been removed from the stable-pose description.

~~The CSO objective now uses the terminal state $\boldsymbol{x}_T$, while the CPO display still uses N in the stage sum and T in its constraints.~~ The CSO objective now uses the terminal state $\boldsymbol{x}_T$ with horizon index $k = 0, \ldots, T-1$, and the CPO formulation consistently uses N in the stage sum, the terminal cost $V_f(\boldsymbol{x}_N)$, and its constraints, so the two horizons are no longer mixed.

The fallback strategy for local-frame construction is present in Eq. (18): when the reference axis a is nearly parallel to n_i, another coordinate axis is selected. The success threshold in Eq. (51) is $1 - (\boldsymbol{\psi}_{\text{goal}}^{\top}\boldsymbol{\psi}_o)^2 \leq 0.015$, corresponding to an orientation tolerance of approximately $14.1°$. The mass is sampled from the discrete set $\{0.01, 0.1, 1.0, 10.0\}$ kg.

Reviewer 5

Q1: Quality

- The authors cover a large span of CI-MPC works from recent literature and appropriately denote inspiration for the proposed surrogate contact model.

- The experimentation is thorough and extensive, each validating the overall efficacy of SCSP across a broad range of tasks.

- The ranking strategy of contact locations, optimized by contact selection optimization (CSO) and sliding window switching strategy as described in Alg. 2, and the various attractive and repulsive potentials, described in Eqns. 41-46, are fairly engineered and require significant heuristic tuning, as briefly described at the bottom of page 11.

- Yet, this ranking strategy is the single most important component of SCSP, smoothing out the volatility of contact selection with some heuristics and yielding superior performance compared to the baseline methods. Since the ranking strategy is such an integral component of SCSP, its ablation and contribution characterization should be elaborated on a good deal more. I am curious how often contact switching, as defined in Eq. (37)-(40) with the grid-swept parameters, is actually triggered. Is there a measure of how helpful it is?

- The authors duly note that the greatly simplified surrogate contact model (SCM) employed for fast CSO limits optimization to point contacts and prevents sequential state prediction due to poor simulation fidelity of complex contact/geometry and deformable objects. This is a large limitation, but broadly endemic to CI-MPC.

- Contribution four claims manipulation "using only vision perception, without the assistance of markers or motion capture systems." I think caveating with the one-time manual MuJoCo calibration is necessary here.

Answer: Thank you for recognizing the importance of the ranking strategy and providing valuable suggestions. We have made the following revisions accordingly.

1) Contribution of ranking strategy:

We now characterize the concrete contribution of the ranking strategy in two complementary ways. First, we evaluate overall task success with ablations. A1 strictly follows the CSO-optimal contact p_r^*, while A2 always selects the nearest contact p_{near}. The SCSP success rate is clearly higher than both monotonic policies in Table II, showing the benefit of switching between the two contact strategies rather than committing to only one.

Second, Section V-B evaluates the value of an individual ranking decision with matched counterfactual rollouts. We define the terminal reward as $r_T = \mathbb{I}(\text{success})$ and the value of a contact-selection decision b at state s_t as

$$J(s_t, b) = \mathbb{E}_{\tau \sim \pi(\cdot|s_t,b)}\left[\gamma^T \mathbb{I}(\text{success})\right], \qquad \gamma = 0.99. \quad (10)$$

We sample SCSP rollout trajectories and identify decision points where p_r^* differs from p_{near}. We then reverse the recorded decision and continue the rollout with the full SCSP controller. For example, if the ranking strategy decides to break the current contact and detour to p_r^*, the counterfactual keeps the current contact; if it decides to maintain contact, the counterfactual forces detachment and detours to p_r^*.

Let J_{RS} and J_{cf} denote the values of the original and reversed decisions. We quantify the decision effect with the normalized advantage

$$\widetilde{A}(s_t) = \begin{cases} \dfrac{J_{\text{RS}} - J_{\text{cf}}}{\max\{J_{\text{RS}}, J_{\text{cf}}\}}, & \max\{J_{\text{RS}}, J_{\text{cf}}\} > 0, \\ 0, & \text{otherwise.} \end{cases} \quad (11)$$

The results are reported in Table VI. We report both the CSO-optimal contact-point switching frequency f_r^* and the physical contact-switching frequency f_{switch}, which quantify how often the ranking strategy is triggered at the guidance and execution levels. We also report the selection probabilities of p_r^* and p_{near}; they are relatively balanced, showing that the ranking strategy makes substantial use of both contact strategies. The normalized advantage $\widetilde{A}$ measures the value improvement contributed by each individual ranking decision.

2)Poor simulation fidelity: Similar to most CIMPC methods, SCM introduces modeling errors due to contact model simplification, and the point-contact assumption further limits its applicability to deformable objects. However, our experiments show that SCSP maintains robust performance under perturbed physical properties in simulation and without system

identification in real-world experiments. This demonstrates that SCSP tolerates single-step simulation errors, which is further supported by our additional analysis experiments. With high-frequency replanning, such simulation errors do not significantly affect the contact selection and planning decisions of SCSP.

SCSP has limitations in highly dynamic tasks and deformable object manipulation. Further extending SCSP may require a higher-fidelity contact simulator or a learned simulator.

3)Clarification of contribution: Although our object pose estimation does not rely on motion capture, we clarify that a one-time calibration estimates only object scale and positional offsets from kinesthetic teaching. The MuJoCo proxy is used for visualization and coarse kinematic alignment; the rendered object pose comes from FoundationPose and no object collision geometry is defined in the proxy.

Q2: Clarity

- The bullet points in the "main contributions" paragraph are mostly rehashed from the preceding paragraph beginning with "In this work, ..." Only one of these large text blocks is necessary.

- Broadly, Section I uses adjective-dense and redundant phrasing, and it should get a revision for clarity and concision.

- pg. 6 Sec. III.B: how are semantically unsuitable regions for manipulation determined? The other heuristics for valid region selection are similarly underexplained.

- Fig. 9b1-4 legend items and axes are hard to read. To accommodate this, the Fig. 9a1-4 panels can have reduced footprint.

- The unit of time in all the x-axis timesteps is not defined, please clarify. I'm assuming it is 0.02s, the value of h?

- I would rename Case Study II from Long-Horizon Tasks to Articulated Object Manipulation. The opening sentence does not make much sense, particularly the phrase "exhibits physical inference abilities." More simply put could be: "SCSP leverages active contact selection to perform robust articulated object manipulation." Additionally, neither Sec. VI.B paragraph, the theoretical formulation of CSO, nor Fig. 18 provide evidence that active CSO, compared to the open-loop CSO, aids in avoiding singularities.

- Referring to CSO as a world model throughout Sec. VII is understandable but also a case of concept creep. The comparison between proposed CSO to PointWorld [55] is a big jump over simply higher-fidelity physical simulation.

Answer: Thank you for the constructive suggestions. We have revised Section I by removing redundant descriptions and improving the clarity and conciseness of the presentation. We have also adjusted the layout of the simulation figures. For the semantically unsuitable regions in Sec. III.B, they are introduced as optional task-level constraints to improve the generality of SCSP. In practice, the mask combines manual annotation with rule-based generation. ~~Manual annotation removes unsuitable contacts: in the soccer task, the left foot is physically unable to~~ Both the manual and the rule-based masks are specified offline together with the object model; no online segmentation or human annotation is performed during manipulation (Section III-B). Manual annotation removes semantically unsuitable regions: in the articulated-object tasks, the object is segmented offline into parts such as handles, and non-manipulable parts are excluded; in the soccer task, points close to the ground ($z_i < 0.02$ m) are masked, and each foot is activated only when the ball lies in its half-space, with workspace constraints imposed on each foot. Rules mask contacts that are inadmissible in the current state: the drawer interior and the microwave-door interior are masked before the drawer or door is pulled open, and from the current end-effector pose we coarsely sample a 3D Cartesian region and mask contact points outside it, since they may require a large change in contact posture. When the contact point on the handle leaves this region, the contact switches to the drawer panel or to the inner wall of the microwave door (Section VI-B). We have added related explanations and clarified that these heuristics are extensibility designs rather than essential components of our experiments.

We agree that the original title "Long-Horizon Tasks" is not sufficiently accurate. The presented experiments demonstrate that SCSP's active contact selection and contact switching strategy can be extended beyond conventional push and flip tasks to improve robustness in daily-life articulated object manipulation scenarios. This further suggests the potential of SCSP as a general contact strategy generator for providing diverse manipulation experiences for future learning-based frameworks. Therefore, we have renamed this case study as "Articulated Object Manipulation" and revised the opening statement accordingly. We have also clarified the role of the ranking strategy in handling robot feasibility issues. Specifically, CSO is object-centric and morphology-agnostic, and thus the optimal contact location $\hat{p}_r^*$ may not consider robot kinematic limitations. SCSP addresses this issue by allowing a nearby feasible candidate p_{near} to replace the infeasible contact prior. We have added this clarification in Sec. VI.B ~~and Fig. 18~~ (the paragraph accompanying Fig. 19 and Table X), and removed the overstatement that CSO explicitly avoids singularities; the revised text attributes the recovery to the ranking strategy rather than to CSO.

Finally, we agree that the term "world model" in Sec. VII is overly broad. Our original intention was to indicate that SCM could potentially be replaced by a learned simulator to extend SCSP to more complex scenarios. Since "world model" has a broader meaning in recent literature, we have replaced it with "simulator" throughout Sec. VII.

Q3: Significance

- The core algorithmic simplification presented in the method, namely diagonalization of the Delassus matrix in the SCM for smooth, efficient optimization in CSO, is heavily inspired, as noted by the authors, by Jin's complementarity-free MPC work [27], which somewhat similarly diagonalizes the Delassus matrix as a scaled identity matrix. Where Jin justifies this extreme physical simplification with the ~~ball, and~~ control, the authors propose a heuristic-driven smoothing over this simplified, open-loop contact selection optimization via

the ranking strategy. Both make fairly strong assumptions about rigid bodies and validate the approach empirically.

- The authors provide finite bounds on the error of this simplification in Lemma 3. However, if I am understanding correctly, the defined bounds are very large: $\sigma_1/\lambda_{\text{min}}$ (Eqn. 58), where λ_{min} is the minimal eigenvalue of the Delassus matrix (W_{env}), which under rank-deficient Jacobians (essentially the default case) is determined by the selected, typically very small regularizing term R in Eq. 22, which can be increased but is itself already a physically spurious quantity. I also would appreciate some extrapolation on "standard perturbation theory for LCP." Generally, I think the authors introduce limited theoretical novelty, but that should not detract from the work's strong and broad results, including real-world deployment and practicality (not needing mocap).

Answer:

Thank you for the careful assessment. We are indeed deeply inspired by Jin's complementarity-free simplification: we construct a closed-form contact model so that CSO can evaluate contact candidates efficiently. Our setting differs from the ComFree formulation in that ComFree requires a tuned matrix parameter K, whereas we design SCM to avoid this additional tuning.

We also need to correct an omission in the previous manuscript. The one-step global projection of the environment impulse was not described. Directly constructing D from the diagonal entries of the Delassus matrix is an aggressive simplification and can introduce substantial model bias, which makes the corresponding error bound loose; the global projection corrects part of this bias. Nevertheless, as the reviewer observes, the bound can still become very large under rank-deficient Jacobians. SCM can represent separation of retained old contacts, but newly created contacts are absent from the current contact set and inevitably introduce additional prediction bias.

For the same reason, the previous lemma and proof are not useful as general guarantees. The earlier appeal to standard perturbation theory for LCPs was also inappropriate: perturbations near a contact-set boundary can cause large changes, and SCM does not smooth those contact-set transitions. We have therefore removed the error-bound lemma and its proof and no longer present this perturbation argument as a theoretical guarantee.

We instead validate SCM empirically as a feasible contact model for contact selection. The velocity-error table in the manuscript compares MuJoCo's high-fidelity QP/LCP contact simulation, ComFree, ContactSDF, and SCM over contact transition, sliding, sticking, and a real flipping rollout, reporting linear- and angular-velocity prediction errors. SCM remains within an acceptable range across these contact states. We then generate optimal contact-wrench trajectories with each model and test their feasibility in the direct-force evaluation. The contact-guidance results show that all models produce largely feasible trajectories, while the closed-form models can perform better in deployment than the higher-fidelity model. This demonstrates that model fidelity is not directly proportional to task success and motivates using SCM for CSO.

Q4: Various typos and minor addenda

Various typos and minor addenda:

- pg. 1: "thus efficiently handle"; "thus efficiently handling"
- pg. 2: linear complementarity system should be defined before the "LCS model" acronym
- I believe [30] and [31] have been conflated into one in the related works and in Table 1.
- pg. 2 "This work inspiring"; "this work is inspiring", or better "Our approach is inspired by this work,"
- pg. 2 "existing CIMPC formulation struggle"; "existing CIMPC formulations struggle"
- pg. 2 "by fast simulating state transitions" ; "by fast simulation of state transitions"
- Eqn. 6 the iterator i should be k to be consistent with following usage
- pg. 6 "the sampling process is performed offline"; what is the computational cost of sampling?
- pg. 13 Floating "I-MPC" after "quaternion error." in the second paragraph

Answer: We corrected the Eq. (6) index and the notation issues that are reflected in the current formulation, retained the explicit offline FPS sampling description, and report its measured offline cost in the computation-time table. The current manuscript also states the $h = 0.02$ s simulation step, the 14.1° orientation tolerance, and the discrete mass distribution used in the experiments.